Sheikh Hasib Cheragee

Lecturer | EEE | AIUB | Dhaka, Bangladesh
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EDUCATION

- **Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh.**
  PhD Fellow in Electrical and Electronic Engineering (Started in April 2024 Semester)
- **Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh.**
  Master of Science (MSc) in Electrical and Electronic Engineering (August 2023)
  CGPA: **3.71 out of 4.00**, Rank: Top 5% in the class | Focus: Electronics and Photonics
- **University of Rajshahi, Rajshahi, Bangladesh.**
  Bachelor of Science (BSc) in Electrical and Electronic Engineering (June 2021)
  CGPA: **3.95 out of 4.00**, Rank: **1st** in a class of 33 students | Focus: Electronics

RESEARCH INTERESTS

Solar Cell; Microelectronics and Nanotechnology; Semiconductor Material and Devices; Wide-bandgap Semiconductor; Power Electronics: Optoelectronics and Photonics: Machine Learning

PROFESSIONAL EXPERIENCE

- **American International University-Bangladesh (AIUB)**
  Lecturer, Department of Electrical and Electronic Engineering 16 October 2024 – Present
  Courses Taught: Introduction to Electrical Circuits, Electronic Device, Electrical Machine-2, Electrical machine-1 Laboratory, Electrical Circuit-2 Laboratory and Electronic Device Laboratory.
- **State University of Bangladesh (SUB)**
  Lecturer, Department of Computer Science and Engineering 01 June 2023 – 15 October 2024
  Courses Taught: Electronic Device and circuits, Digital Logic Design, Electrical circuit analysis, Electronic Device and circuits Laboratory, Digital Logic Design Laboratory, Electrical circuit analysis Laboratory.
- **Bangladesh University of Engineering and Technology(BUET)** 01 October 2021 – 01 April 2023
  Research Fellow, Department of Electrical and Electronic Engineering

PUBLICATIONS

Journal Articles

1. **Sheikh Hasib Cheragee** and Mohammad Jahangir Alam, "Device modeling and numerical study of a double absorber solar cell using a variety of electron transport materials," *Heliyon*, Volume: 9, Issue No. 7, 2023, pp. 1-11, https://doi.org/10.1016/j.heliyon.2023.e18265
2. **Sheikh Hasib Cheragee**, and Mohammad Jahangir Alam, "Device modelling and numerical analysis of high-efficiency double absorbers solar cells with diverse transport layer materials." Results in Optics 15 (2024): pp. 100647, https://doi.org/10.1016/j.rio.2024.100647
3. **Sheikh Hasib Cheragee**, Jannatul Afroj Akhi, Mishion Dev, Md Mahfuzul Haque, and Mohammad Jahangir Alam, "Numerical Design of Non-toxic High-Efficiency Tandem solar cell with distinct hole transport Materials." Thin Solid Films (2024): pp. 140432, https://doi.org/10.1016/j.tsf.2024.140432
4. Md Alamgir Hossain, Tahmid Hasan, Fahad Ahmed, **Sheikh Hasib Cheragee**, Muntasir Hasan Kanchan, and Md Alimul Haque. "Towards Superior Android Ransomware Detection: An Ensemble Machine Learning Perspective." Cyber Security and Applications (2024), pp.100076, https://doi.org/10.1016/j.csa.2024.100076
5. **Sheikh Hasib Cheragee**, Nazmul Hassan, Sakil Ahammed and A. Z. M. T. Islam "A study of IoT-based real-time solar power remote monitoring system " International Journal of Ambient Systems and Applications, ISSN: 2320-9259, Volume: 9, Issue No. 1/2, 2021, page. 27-36, doi: 10.5121/ijasa.2021.9204
6. Sakil Ahammed, Nazmul Hassan, **Sheikh Hasib Cheragee**, and A. Z. M. T. Islam "An IoT-based real-

time remote health monitoring system" International Journal of Recent Engineering Science, ISSN: 2349 –7157, Volume: 8, Issue No. 3, 2021, page. 23-29, doi:10.14445/23497157/IJRES-V8I3P104

7. Nazmul Hassan, **Sheikh Hasib Cheragee**, Sakil Ahammed and A. Z. M. T. Islam "Sensor based Smart Irrigation System with Monitoring and Controlling using Internet of Things" International Journal of Ambient Systems and Applications, ISSN: 2320-9259, Volume: 9, Issue No. 1/2, 2021, page. 17-26, doi: 10.5121/ijasa.2021.9203

Conference Articles

1. **Sheikh Hasib Cheragee** and M. J. Alam, "2-T Non-Toxic Tandem solar cell with Distinct HTL Materials," 2022 12th International Conference on Electrical and Computer Engineering (ICECE), Dhaka, Bangladesh, 2022, IEEE Xplore, pp. 152-155, doi: 10.1109/ICECE57408.2022.10089070.
2. **Sheikh Hasib Cheragee**, A. Z. M. T. Islam and M. J. Alam, "2-T Non-Toxic Multijunction solar cell with Diverse ETL materials," 2022 International Conference on Recent Progresses in Science, Engineering and Technology (ICRPSET), Rajshahi, Bangladesh, 2022, IEEE Xplore, pp. 1-4, doi: 10.1109/ICRPSET57982.2022.10188555.
3. M. M. Haque and **Sheikh Hasib Cheragee**, "Performance Analysis of n+ Pocket DG-JLFET with SiO_2 and High-K Oxide Thin Layer for Biomedical Application," 2022 International Conference on Innovations in Science, Engineering and Technology (ICISET), Chittagong, Bangladesh, 2022, IEEE Xplore, pp. 290-294, doi:10.1109/ICISET54810.2022.9775872.
4. S. C. Pall, **Sheikh Hasib Cheragee**, et al., "Performance Analysis of Machine Learning and ANN for Load Forecasting in Bangladesh," 2023 International Conference on Next-Generation Computing, IoT and Machine Learning (NCIM), Gazipur, Bangladesh, 2023, IEEE Xplore, pp. 1-5, doi: 10.1109/NCIM59001.2023.10212451.
5. **Sheikh Hasib Cheragee** et al., "Short- and Medium-Term Electrical Load Forecasting in Bangladesh Using Machine Learning Approaches," 2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka, Bangladesh, 2024, IEEE Xplore, pp. 1-5, doi: 10.1109/iCACCESS61735.2024.10499506.
6. M. J. Islam, **Sheikh Hasib Cheragee**, M. M. Tarek and F. Gulshan, "A numerical analysis of a high-efficiency solar cell using a DFT explored nontoxic perovskite material," 2024 3rd International Conference on Advancement in Electrical and Electronic Engineering (ICAEEE), Gazipur, Bangladesh, 2024, IEEE Xplore, pp. 1-5, doi: 10.1109/ICAEEE62219.2024.10561832.
7. M. J. Islam, **Sheikh Hasib Cheragee**, M. A. Haque Chowdhury, S. Subrina, M. Hossain and F. Gulshan, "Numerical Study of Higher Efficiency Perovskite Solar Cell Using Density Functional Theory," 2024 International Conference on Advances in Computing, Communication, Electrical, and Smart Systems (iCACCESS), Dhaka, Bangladesh, 2024, IEEE Xplore, pp. 1-5, doi: 10.1109/iCACCESS61735.2024.10499519.
8. Selim Molla, **Sheikh Hasib Cheragee** et al., " Potential Charging Techniques for Electric Vehicles Utilizing Photovoltaic Sources in Bangladesh," 2024 International Conference on Power, Electrical, Electronics and Industrial Applications (PEEIACON), IEEE Xplore, Rajshahi, Bangladesh, 2024 (Accepted).
9. **Sheikh Hasib Cheragee**, Jannatul Afroj Akhi, Mohammad Tahsin Alam, and Mohammad Jahangir Alam, " Modeling and Performance Investigation of Perovskite/$MoTe_2$ solar cell exceeding 44% Efficiency: A SCAPS-1D Simulation" 2024 International Conference on Recent Progresses in Science, Engineering and Technology, IEEE Xplore, Rajshahi, Bangladesh, 2024 (Accepted).
10. M. T. Rahman, S. Chandra Pall, R. B. Rashid, M. Shifat Hossen, S. K. Das and **Sheikh Hasib Cheragee**, "SolarNet- An efficient multi-model ensemble approach for Solar Power Forecasting based on Artificial Intelligence," 2023 6th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2023, IEEE Xplore, pp. 1-6, doi: 10.1109/EICT61409.2023.10427929.
11. Anisuzzaman, N. Sarker, **Sheikh Hasib Cheragee** and M. Niamul Hasan, "Design of Microstrip Patch Antenna With Improved Return Loss and Bandwidth for 5G Wireless Communication," 2023 6th International Conference on Electrical Information and Communication Technology (EICT), Khulna, Bangladesh, 2023, IEEE Xplore, pp. 1-4, doi: 10.1109/EICT61409.2023.10427863.

AWARDS AND ACHIVEMENTS

- Received the "**Bangabandhu Gold Medal Award**" in recognition of securing the First position in the Faculty of Engineering in the B.Sc. Engineering examination of the year 2019 from the dept. of EEE, University of Rajshahi.
- Nominated for the "**Prime Minister Gold Medal Award- 2021**" for obtaining the **Highest CGPA** in BSc Engg. among all the students of the Faculty of Engineering, University of Rajshahi.
- Received the "**Post-Graduation Fellowship"** at BUET from 01 October 2021 to 01 April 2023 (A total of 1 year and 6 months).
- Received the **"Dean's Award**" in the consecutive academic years of 2016-2017, 2017-2018, 2018-2019, and 2019-2020 from the Faculty of Engineering, University of Rajshahi, Bangladesh for Excellent Academic standing.
- Received **APEE Reunion 2011 Award**" in **"Two times"** for outstanding result in BSc Engg. Part-3 and Part-4 examinations.
- Received a **Scholarship for 7 days** From November 18, 2019, to November 24, 2019, for joining an invitation program carried out under the framework of the **"Sakura Science Exchange Program"** administered by the **Japan Science and Technology Agency (JST), Tokyo, Japan.**

TECHNICAL SKILLS

- **Programming Languages:** Python, Matlab, C, Arduino.
- **Simulation Software:** Matlab, Simulink, SCAPS 1D, Sentaurus TCAD Simulation, Materials Studio, Multisim.
- **Graphics:** Adobe Photoshop, Adobe Illustrator.
- **Computer Skills:** Microsoft Office, Latex.

STANDARDIZED TEST SCORE

TOEFL: 99 (Listening: 28; Reading 26; Writing 27; Speaking:18)

LEADERSHIP AND CO-CURRICULAR ACTIVITIES

BUET Post-graduation Research Alliance
Vice-President *February 2022 - March 2023*

- **Research Conclave 2022 (Virtual):** Organized the virtual seminar on research methodology among all the members.
- **Games:** Participated in PGRA Shahid Smriti Hall Games 2022 and got runner-up in Badminton.

IEEE Rajshahi University Student Chapter
Founding Member

- **Research and Innovation Conclave 2021 (Virtual)**: Organized the virtual seminar on the implications of silicon and GaN material in the semiconductor device.

Rajshahi University EEE Association
General Secretary *June 2019 - May 2020*

- **Electronics Workshop**: Organized a two-day-long electronics workshop for the fresher students on basic electronics and their diverse application in practical devices.

REFERENCES

Dr. Mohammad Jahangir Alam Professor, Department of Electrical and Electronic Engineering, Bangladesh University of Engineering and Technology Dhaka-1000, Bangladesh. Email: mjalam@eee.buet.ac.bd Phone: 01911356905	**Dr. Abu Zafor Md. Touhidul Islam** Professor and Ex- Chairman, Department of Electrical and Electronic Engineering University of Rajshahi Rajshahi-6205, Bangladesh. Email: touhid.eee@ru.ac.bd Phone: 01733788812